

Introduction to Visual Data Analytics DV1699 – VT2026

Multiphase Summative Assessment (3 credits)

Deadlines: according to Canvas

General Description

This multiphase summative assessment is an assignment consisting of multiple parts. Its focus is on applying the different concepts and techniques presented during the labs.

The first two parts (Part I and Part II) are written assignments in which you will practice programming using Python visualization libraries and answer some questions related to the analysis of datasets using visualization. Part I covers Lab 1 (Data Collecting, Processing, and Plotting) while Part II covers Lab 2 (Interaction techniques).

Part III consists of an oral presentation of the work submitted for Parts I and II.

The detailed description of the tasks for Part I and Part II will be published separately, each part right before its corresponding lab lecture.

Although discussion among students is encouraged in the course, the implementation you submit must be done personally by you. Any found act of plagiarism will be reported and issued to the BTH Disciplinary Committee. For more details, see the following link:

<https://www.bth.se/english/student/during-your-studies/rules-and-ethics>

General Instructions

Completion of the whole assignment consists of (1) submitting code and text answers for BOTH Part I and Part II using Jupyter notebook (. *ipynb* format) before the chosen deadline, and (2) presenting and discussing the proposed solutions during an oral presentation that will be performed after the submission deadline. The presentation time slots will be communicated in Canvas after the deadline.

The tasks in Parts I and II total 100 points: 50 points for Part I and 50 for Part II.

For both Part I and Part II, a compulsory task of 25 points must be completed correctly for an E grade. Additional tasks are available in each Part to reach a higher grade (D-A) upon corrected completion.

The submission will be graded on a scale of A (90+ points), B (80-89 points), C (70-79 points), D (60-69 points), E (50-59 points), FX (45-49 points, failed result with a little more work required), and F (<45 points, failed result).

Keep in mind that also for grades higher than E, the 2 compulsory tasks must be completed correctly.

After the submission, students can book a presentation timeslot. A presentation will be graded as G (godkänd/pass) or U (underkänd/fail). For a passing grade (G), during the presentation and discussion, the student must show a good and complete understanding of

the code they implemented, making at most only one conceptual error related to the theory applied in the code. If these conditions are not met the whole examination will be graded as F (even if the points from the submission are ≥ 50).

The final passing grade is based on passing grades in both Submission and Presentation.

In case of Presentation graded G and Submission graded FX, the student will have 2 weeks to fix the small problems in the code highlighted in the grading feedback and resubmit the Jupiter notebook submission. Once the resubmission is complete, the teacher will inspect it. If the problems noted in the feedback are fixed and no new problems are introduced, the submission will be marked as pass (E); otherwise, it will be graded F and the entire examination will be graded F. Remember that, according to BTH rules, if a submission is graded FX, it can only be complemented up to an E grade.

A brief overview of the grading component mapping is shown in Table 1.

Table 1. Grading component mapping.

Notebook Submission		Presentation
Part I	Part II	Part III
CT1 (25 points)	CT2 (25 points)	G
T1 (12.5 points)	T3 (8 points)	
T2 (12.5 points)	T4 (17 points)	

	Compulsory elements for grade E
	Selective elements for a grade higher than E

Submission Files Instructions

The submission for this examination should be a .zip file that includes your single Jupyter notebook (.ipynb) containing the code and answers for both Part I and Part II, any additional datasets required by the notebook, and/or generated .html files or figures. The submission must be re-executable, and the grading is based on the re-executed outputs.

Specifically, the .ipynb file should include:

- Student's name.
- The proposed answer to each of the tasks from the assignment (Part I and Part II).
- A justification for each of the provided answers, together with all the required visualizations to back it up.

Additionally, despite the correctness of answers, all visualizations in the submission should:

- Have a title and respective axes labels.
- Have descriptive legend names: instead of leaving the variable names or the function syntax as a legend, each plot function should be labeled after what it is visualizing.
- Present figures in a size that preserves readability. E.g., a figure size around `figsize= (14, 7)` is recommended.